

Martinez, Thomas - Education and Background Statement

I was born and raised in Miami, Florida, by Brazilian immigrant parents. I attended public schools, beginning in a Portuguese bilingual program and later joining a high school of more than 3,000 students. With most of my family living in Brazil, I found a second family in my school communities. That sense of belonging has shaped what I value in education and the communities I want to help build as a mathematician.

I entered Harvey Mudd College intending to study computer science, but a discrete mathematics course taught by Arthur Benjamin drew me toward combinatorics. Alongside my studies, I served in student government and became an Academic Excellence mathematics tutor, supporting first- and second-year students through the college's Core curriculum. When the pandemic interrupted campus life, we continued tutoring virtually. The mathematics department recognized my contributions with the Alvin White Prize, which honors the humanistic side of mathematics. Tutoring gave me an early model for how students can take responsibility for one another's education.

After graduating in 2021, I returned to Miami as a City Year AmeriCorps member to support students returning to in-person learning. As Team Leader at Miami Jackson Senior High School, I taught four Algebra I classes for students at risk of not graduating on time. More than half had recently immigrated from countries including Honduras, El Salvador, and Cuba, and I taught them mathematics in Spanish. I also designed weekly interventions for students needing additional support and volunteered as a tutor for elementary and middle-school students at The Motivational Edge. This work made me more attentive to the circumstances surrounding a student's performance: language, disrupted schooling, and access to individual attention all mattered to how I approached teaching.

At UCLA, I continued to make collaborative learning part of my own education. While preparing for the Geometry/Topology qualifying exam, I organized a study group. Together, we wrote a study guide of more than 100 pages covering key topics and solutions to eight previous exams, creating a resource for future students as well as ourselves. I have also helped build mathematical community by co-organizing the 21st International Fibonacci Conference at Harvey Mudd College in July 2024 and organizing UCLA's Combinatorics Research Seminar.

Through UCLA's Directed Reading Program, I have mentored eight undergraduates on topics ranging from graph theory to tiling proofs of Fibonacci identities. With Olha Shevchenko and three of my undergraduate mentees, I recently submitted a paper on q -Fibonacci combinatorics to a journal. Working with students as they move from learning established mathematics to contributing to research has become an important part of the academic career I want to pursue. As a program committee member, I have also recruited graduate mentors and helped streamline participation for mentors and mentees.

My work with UCLA's Learning Assistant program has extended this commitment to introductory courses. Learning Assistants are undergraduate peer educators who help graduate teaching assistants incorporate active learning into discussion sections. Having worked with LAs in my own sections, I served as mathematics LA coordinator, connecting instructors with the program and helping extend its reach to additional courses. UCLA's mathematics department recognized my teaching and service with the Raymond Redheffer Prize. This role taught me to support student learning through both classroom teaching and the coordination that makes peer instruction possible.

As a postdoctoral fellow at UC Davis, I would seek to mentor undergraduate research through the Davis Math Lab, developing combinatorics projects with accessible starting points and room for sustained investigation. As a faculty member, I hope to establish a summer Research Experiences for Undergraduates program involving graduate students and faculty as mentors. I also intend to continue supporting peer instruction in introductory mathematics. My experiences in Miami, at Harvey Mudd, and at UCLA have given me concrete ways to contribute to the university's public service mission: helping students persist in foundational courses, sharing resources, and opening opportunities to participate in research. I want students to find in mathematics both intellectual challenges and a community that supports their growth.